

Assignment 1 – SPA
Situated Learning Assignment
Domain- Smart Cities & Urban Infrastructure

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TASK B - Apache Kafka — Multi-Source Urban Data Ingestion

Problem Statement 4:

Set up a 3-broker Kafka cluster. Create topics for all four streams with justified partition counts. Implement topic-level retention policies: bus GPS (retain 24 hours — for replay during investigation of accidents), air quality (retain 90 days — for pollution trend analysis), smart meters (retain 365 days — for regulatory energy audits). Document and justify each retention choice.

Solution 4:

Kafka Cluster Design

Cluster Setup

Broker 1 : kafka-01

Broker 2 : kafka-02

Broker 3 : kafka-03

Replication Factor = 3

Topic Configuration

Topic	Rate	Partitions	Retention
urbanpulse.bus_gps	2400/sec	24	24 Hours
urbanpulse.traffic_signals	380/sec	8	30 Days
urbanpulse.air_quality	60/sec	4	90 Days
urbanpulse.smart_meters	1100/sec	12	365 Days

Retention Justification

Bus GPS (24 Hours)

- Accident investigations
- Traffic incident replay
- Reduced storage cost

Air Quality (90 Days)

- Seasonal pollution patterns
- Environmental audits

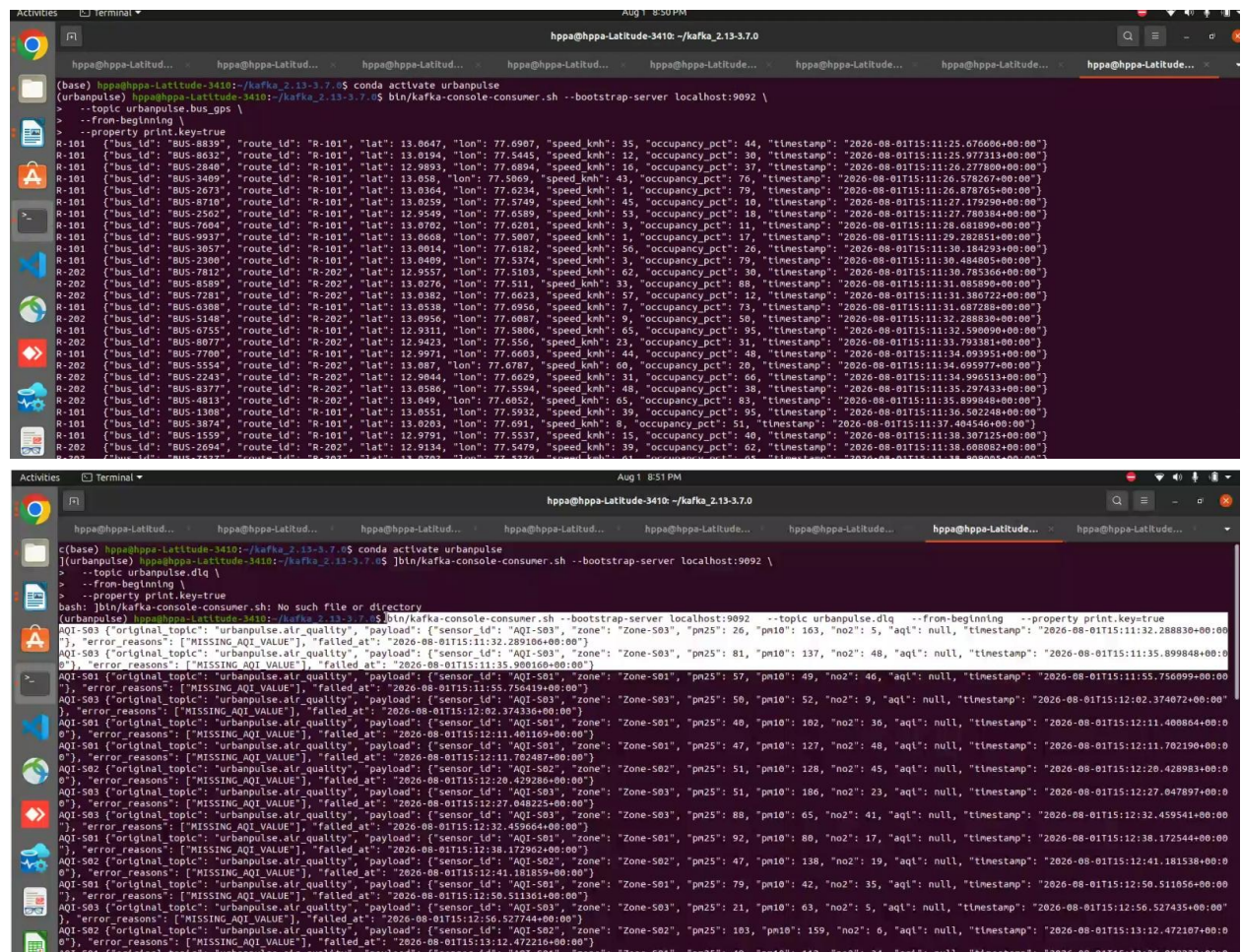
Smart Meters (365 Days)

- Regulatory compliance
- Annual energy audits

Problem Statement 5:

Implement Python producers for the bus_gps and air_quality streams. The air_quality producer must implement at-least-once semantics with retry logic (sensors occasionally timeout). The bus_gps producer must use a key of route_id to guarantee ordering of bus positions per route. Embed a simulated sensor failure: 5% of air quality events arrive with null AQI values — your producer must handle and log these gracefully.

Solution 5:



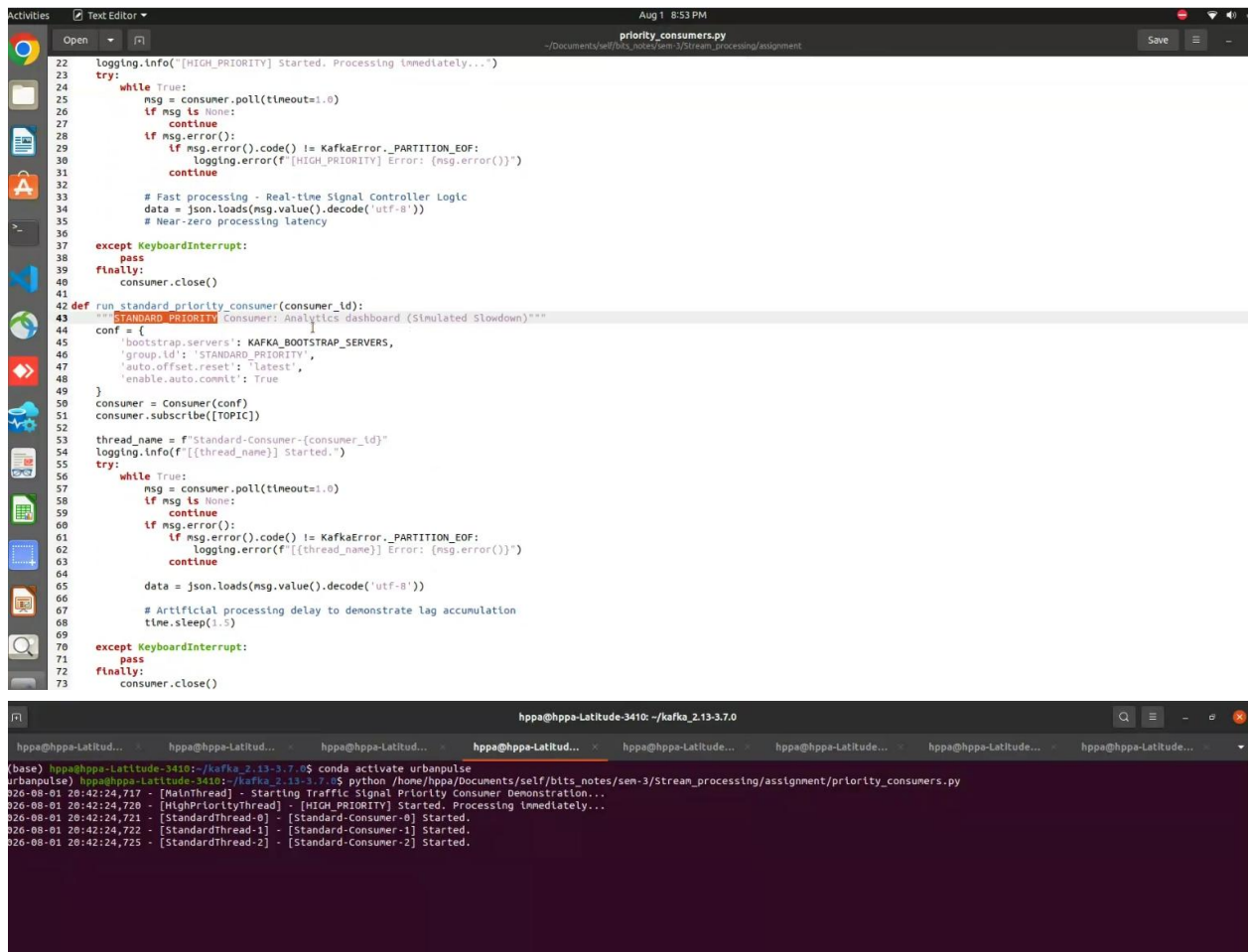
```
(base) hppa@hppa-Latitude-3410:~/kafka_2.13-3.7.0$ conda activate urbanpulse
(urbanpulse) hppa@hppa-Latitude-3410:~/kafka_2.13-3.7.0$ bin/kafka-console-consumer.sh --bootstrap-server localhost:9092 \
> --topic urbanpulse.bus_gps \
> --from-beginning \
> --property print.key=true
R-101 {"bus_id": "BUS-8839", "route_id": "R-101", "lat": 13.0647, "lon": 77.6907, "speed_kmh": 35, "occupancy_pct": 44, "timestamp": "2026-08-01T15:11:25.676060+00:00"}
R-101 {"bus_id": "BUS-8832", "route_id": "R-101", "lat": 13.0194, "lon": 77.5445, "speed_kmh": 12, "occupancy_pct": 30, "timestamp": "2026-08-01T15:11:25.977313+00:00"}
R-101 {"bus_id": "BUS-2840", "route_id": "R-101", "lat": 12.9893, "lon": 77.6894, "speed_kmh": 16, "occupancy_pct": 37, "timestamp": "2026-08-01T15:11:26.278000+00:00"}
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R-101 {"bus_id": "BUS-2673", "route_id": "R-101", "lat": 13.0364, "lon": 77.6234, "speed_kmh": 1, "occupancy_pct": 79, "timestamp": "2026-08-01T15:11:26.678765+00:00"}
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c(base) hppa@hppa-Latitude-3410:~/kafka_2.13-3.7.0$ conda activate urbanpulse
(urbanpulse) hppa@hppa-Latitude-3410:~/kafka_2.13-3.7.0$ bin/kafka-console-consumer.sh --bootstrap-server localhost:9092 \
> --topic urbanpulse.aqi \
> --from-beginning \
> --property print.key=true
bash: bin/kafka-console-consumer.sh: No such file or directory
(urbanpulse) hppa@hppa-Latitude-3410:~/kafka_2.13-3.7.0$ bin/kafka-console-consumer.sh --bootstrap-server localhost:9092 --topic urbanpulse.aqi --from-beginning --property print.key=true
AQI-503 {"original_topic": "urbanpulse.air_quality", "payload": {"sensor_id": "AQI-503", "zone": "Zone-S03", "pm25": 26, "pm10": 163, "no2": 5, "aqi": null, "timestamp": "2026-08-01T15:11:32.288830+00:00"}, "error_reasons": ["MISSING_AQI_VALUE"], "failed_at": "2026-08-01T15:11:32.288106+00:00"}
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```

Problem Statement 6:

Build a priority consumer architecture: implement two consumer groups for the traffic_signals topic — a HIGH_PRIORITY group (1 consumer, reads from all partitions) for the real-time signal control system, and a STANDARD_PRIORITY group (3 consumers) for the analytics dashboard. Demonstrate that the HIGH_PRIORITY group maintains near-zero lag even when the STANDARD_PRIORITY group falls behind during a simulated processing slowdown.

Solution 6 :



The screenshot displays a Linux desktop environment. The top panel shows the date and time as 'Aug 1 8:53 PM'. The main workspace is divided into two windows. The upper window is a text editor titled 'priority_consumers.py' located at '~Documents/self/bits_notes/sem-3/Stream_processing/assignment'. It contains Python code for a Kafka consumer application. The code includes a main function that runs a standard priority consumer, a configuration dictionary for Kafka, and two separate consumer threads. The lower window is a terminal titled 'hppa@hppa-Latitude-3410: ~/kafka_2.13-3.7.0'. It shows the command 'python /home/hppa/Documents/self/bits_notes/sem-3/Stream_processing/assignment/priority_consumers.py' being executed. The terminal output indicates that the application has started successfully, with messages from the main thread and two standard consumer threads.

```
22 logging.info(f"[HIGH_PRIORITY] Started. Processing immediately...")
23 try:
24     while True:
25         msg = consumer.poll(timeout=1.0)
26         if msg is None:
27             continue
28         if msg.error():
29             if msg.error().code() != KafkaError._PARTITION_EOF:
30                 logging.error(f"[HIGH_PRIORITY] Error: {msg.error()}")
31             continue
32         # Fast processing - Real-time Signal Controller Logic
33         data = json.loads(msg.value().decode('utf-8'))
34         # Near-zero processing latency
35
36     except KeyboardInterrupt:
37         pass
38     finally:
39         consumer.close()
40
41
42 def run_standard_priority_consumer(consumer_id):
43     """STANDARD_PRIORITY Consumer: Analytics dashboard (Simulated Slowdown)"""
44     conf = {
45         'bootstrap.servers': KAFKA_BOOTSTRAP_SERVERS,
46         'group.id': 'STANDARD_PRIORITY',
47         'auto.offset.reset': 'Latest',
48         'enable.auto.commit': True
49     }
50     consumer = Consumer(conf)
51     consumer.subscribe([TOPIC])
52
53     thread_name = f"Standard-Consumer-{consumer_id}"
54     logging.info(f"[{thread_name}] Started.")
55     try:
56         while True:
57             msg = consumer.poll(timeout=1.0)
58             if msg is None:
59                 continue
60             if msg.error():
61                 if msg.error().code() != KafkaError._PARTITION_EOF:
62                     logging.error(f"[{thread_name}] Error: {msg.error()}")
63                 continue
64             data = json.loads(msg.value().decode('utf-8'))
65             # Artificial processing delay to demonstrate lag accumulation
66             time.sleep(1.5)
67
68     except KeyboardInterrupt:
69         pass
70     finally:
71         consumer.close()
72
73
```

```
hppa@hppa-Latitude-3410: ~/kafka_2.13-3.7.0
(base) hppa@hppa-Latitude-3410:~/kafka_2.13-3.7.0$ conda activate urbanpulse
urbanpulse) hppa@hppa-Latitude-3410:~/kafka_2.13-3.7.0$ python /home/hppa/Documents/self/bits_notes/sem-3/Stream_processing/assignment/priority_consumers.py
026-08-01 20:42:24,717 - [Mainthread] - Starting Traffic Signal Priority Consumer Demonstration...
026-08-01 20:42:24,720 - [HighPriorityThread] - [HIGH_PRIORITY] Started. Processing immediately...
026-08-01 20:42:24,721 - [StandardThread-0] - [Standard-Consumer-0] Started.
026-08-01 20:42:24,722 - [StandardThread-1] - [Standard-Consumer-1] Started.
026-08-01 20:42:24,725 - [StandardThread-2] - [Standard-Consumer-2] Started.
```

Problem 7:

Build a Kafka Streams application that performs real-time enrichment: join the `bus_gps` stream with a static `route_schedule` KTable (provided as CSV on eLearn) to produce an enriched stream containing `scheduled_arrival_time`, `route_name`, and terminal alongside GPS position — this is the foundation for the real-time ETA service.

Solution 7:


```
priority_consumers.py
1 import time
2 import json
3 import random
4 import logging
5 from datetime import datetime, timezone
6 from confluent_kafka import Producer
7
8 logging.basicConfig(level=logging.INFO, format='%(asctime)s - %(levelname)s - %(message)s')
9 KAFKA_BOOTSTRAP_SERVERS = 'localhost:9092'
10
11 def delivery_report(err, msg):
12     if err is not None:
13         logging.error(f'Delivery failed for record {msg.key()}: {err}')
14
15 # Task B Mandate: At-Least-Once + Idempotent Ingestion Configuration
16 producer_config = {
17     'bootstrap.servers': KAFKA_BOOTSTRAP_SERVERS,
18     'enable.idempotence': True,
19     'acks': 'all',
20     'retries': 5,
21     'max.in.flight.requests.per.connection': 1
22 }
23 producer = Producer(producer_config)
24
25 def get_partition_key(topic, payload):
26     """
27     Task B Constraint Mapping:
28     - bus_gps MUST use route_id to guarantee partition ordering per route.
29     - air_quality uses sensor_id.
30     - smart_meters uses ward_id.
31     - traffic_signals uses junction_id.
32     """
33     if topic == 'urbanpulse.bus_gps':
34         return str(payload.get('route_id', 'UNKNOWN'))
35     elif topic == 'urbanpulse.air_quality':
36         return str(payload.get('sensor_id', 'UNKNOWN'))
37     elif topic == 'urbanpulse.smart_meters':
38         return str(payload.get('ward_id', 'UNKNOWN'))
39     elif topic == 'urbanpulse.traffic_signals':
40         return str(payload.get('junction_id', 'UNKNOWN'))
41     return "UNKNOWN"
42
43 def validate_and_route(topic, payload):
44     """
45     Validates messages against city rules.
46     Routes to original topic if valid, or urbanpulse.dlq if faulty.
47     """
48
49 hppa@hppa-Latitude... hppa@hppa-Latitude... hppa@hppa-Latitude... hppa@hppa-Latitude... hppa@hppa-Latitude... hppa@hppa-Latitude... hppa@hppa-Latitude...
(base) hppa@hppa-Latitude-3410:~/kafka_2.13-3.7.0$ conda activate urbanpulse
(urbanpulse) hppa@hppa-Latitude-3410:~/kafka_2.13-3.7.0$ bin/kafka-console-consumer.sh --bootstrap-server localhost:9092 \
> --topic urbanpulse.bus_gps \
> --from-beginning \
> --property print.key=true
R-101 {"bus_id": "BUS-8839", "route_id": "R-101", "lat": 13.0647, "lon": 77.6907, "speed_kmh": 35, "occupancy_pct": 44, "timestamp": "2026-08-01T15:11:25.676006+00:00"}
R-101 {"bus_id": "BUS-8632", "route_id": "R-101", "lat": 13.0194, "lon": 77.5445, "speed_kmh": 12, "occupancy_pct": 30, "timestamp": "2026-08-01T15:11:25.977313+00:00"}
R-101 {"bus_id": "BUS-2840", "route_id": "R-101", "lat": 12.9893, "lon": 77.6894, "speed_kmh": 10, "occupancy_pct": 37, "timestamp": "2026-08-01T15:11:26.277800+00:00"}
R-101 {"bus_id": "BUS-3409", "route_id": "R-101", "lat": 13.058, "lon": 77.5069, "speed_kmh": 43, "occupancy_pct": 76, "timestamp": "2026-08-01T15:11:26.578267+00:00"}
R-101 {"bus_id": "BUS-2073", "route_id": "R-101", "lat": 13.0364, "lon": 77.6234, "speed_kmh": 1, "occupancy_pct": 79, "timestamp": "2026-08-01T15:11:26.878765+00:00"}
R-101 {"bus_id": "BUS-8710", "route_id": "R-101", "lat": 13.0259, "lon": 77.5749, "speed_kmh": 45, "occupancy_pct": 10, "timestamp": "2026-08-01T15:11:27.179290+00:00"}
R-101 {"bus_id": "BUS-2502", "route_id": "R-101", "lat": 12.9549, "lon": 77.6589, "speed_kmh": 53, "occupancy_pct": 18, "timestamp": "2026-08-01T15:11:27.480384+00:00"}
R-101 {"bus_id": "BUS-7604", "route_id": "R-101", "lat": 13.0702, "lon": 77.6201, "speed_kmh": 3, "occupancy_pct": 11, "timestamp": "2026-08-01T15:11:28.681890+00:00"}
R-101 {"bus_id": "BUS-9937", "route_id": "R-101", "lat": 13.0668, "lon": 77.5807, "speed_kmh": 1, "occupancy_pct": 17, "timestamp": "2026-08-01T15:11:29.282851+00:00"}
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R-101 {"bus_id": "BUS-1494", "route_id": "R-101", "lat": 12.971, "lon": 77.5579, "speed_kmh": 60, "occupancy_pct": 80, "timestamp": "2026-08-01T15:11:42.521101+00:00"}
R-101 {"bus_id": "BUS-4288", "route_id": "R-101", "lat": 13.0872, "lon": 77.5613, "speed_kmh": 5, "occupancy_pct": 57, "timestamp": "2026-08-01T15:11:43.121500+00:00"}
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R-202 {"bus_id": "BUS-4127", "route_id": "R-202", "lat": 13.0132, "lon": 77.6409, "speed_kmh": 14, "occupancy_pct": 40, "timestamp": "2026-08-01T15:11:43.722798+00:00"}
```

Problem Statement 8:

Implement a dead-letter queue (DLQ) pattern: messages that fail validation (null values, out-of-range AQI, impossible GPS coordinates) are routed to a separate urbanpulse.dlq topic with an error reason field. Submit a 5-minute DLQ report showing error type distribution.

Solution 8:

Dead Letter Queue

Validation Rules

1. Null AQI
2. AQI > 500
3. Latitude > 90 or < -90
4. Longitude > 180 or < -180
5. Missing timestamp

```

priority_consumers.py
producers.py

54 errors.append('MISSING_AQI_VALUE')
55 elif aqi < 0 or aqi > 500:
56     errors.append('AQI_OUT_OF_BOUNDS')
57
58 elif topic == 'urbanpulse.bus_gps':
59     lat = payload.get('lat')
60     lon = payload.get('lon')
61     if lat is None or lon is None or not (12.0 <= lat <= 14.0) or not (76.0 <= lon <= 78.0):
62         errors.append('MALFORMED_GEOSPATIAL_COORDINATES')
63
64 key_str = get_partition_key(topic, payload)
65
66 # 2. Routing Logic
67 if errors:
68     # Route to Dead Letter Queue (DLQ)
69     dlq_payload = {
70         'original_topic': topic,
71         'payload': payload,
72         'error_reasons': errors,
73         'failed_at': datetime.now(tz=timezone.utc).isoformat()
74     }
75     producer.produce(
76         topic='urbanpulse.dlq',
77         key=key_str.encode('utf-8'),
78         value=json.dumps(dlq_payload).encode('utf-8'),
79         callback=delivery_report
80     )
81     return False
82 else:
83     # Route to original destination topic with correct key ordering
84     producer.produce(
85         topic=topic,
86         key=key_str.encode('utf-8'),
87         value=json.dumps(payload).encode('utf-8'),
88         callback=delivery_report
89     )
90     return True
91
92 def simulate_urban_pulse():
93     routes = ['R-101', 'R-202', 'R-303']
94     sensors = ['AQI-S01', 'AQI-S02', 'AQI-S03']
95     wards = ['Ward-Alpha', 'Ward-Beta', 'Ward-Gamma']
96     junctions = ['JNC-01', 'JNC-02', 'JNC-03']
97
98     logging.info("Starting validated UrbanPulse multi-source pipeline...")
99
100     try:
101         while True:
102             timestamp = datetime.now(tz=timezone.utc).isoformat()
103             ...

```

```

hppa@hppa-Latitude-3410: ~/kafka_2.13-3.7.0
hppa@hppa-Latit... hppa@hppa-Latit... hppa@hppa-Latit... hppa@hppa-Latit... hppa@hppa-Latit... hppa@hppa-Latit... hppa@hppa-Latit... hppa@hppa-Latit... hppa@hppa-Latit...
(base) hppa@hppa-Latitude-3410: ~/kafka_2.13-3.7.0$ conda activate urbanpulse
urbanpulse) hppa@hppa-Latitude-3410: ~/kafka_2.13-3.7.0$ python /home/hppa/documents/self/blts_notes/sen-3/Stream_processing/assignment/producers.py
026-08-01 20:41:25.676 - INFO - Starting validated UrbanPulse multi-source pipeline...
41785597086.191[GETPID]rdkafka:producer-1 [thrd:main]: Failed to acquire idempotence PID from broker localhost:9092/1: Broker: Coordinator load in progress: retrying
026-08-01 20:41:32.289 - WARNING - Simulating Sensor Outage for AQI-S03: Injected Null AQI routed to DLQ
026-08-01 20:41:35.900 - WARNING - Simulating Sensor Outage for AQI-S03: Injected Null AQI routed to DLQ
026-08-01 20:41:55.756 - WARNING - Simulating Sensor Outage for AQI-S01: Injected Null AQI routed to DLQ
026-08-01 20:42:02.374 - WARNING - Simulating Sensor Outage for AQI-S03: Injected Null AQI routed to DLQ
026-08-01 20:42:11.401 - WARNING - Simulating Sensor Outage for AQI-S01: Injected Null AQI routed to DLQ
026-08-01 20:42:11.702 - WARNING - Simulating Sensor Outage for AQI-S01: Injected Null AQI routed to DLQ
026-08-01 20:42:26.409 - WARNING - Simulating Sensor Outage for AQI-S02: Injected Null AQI routed to DLQ
026-08-01 20:42:27.048 - WARNING - Simulating Sensor Outage for AQI-S03: Injected Null AQI routed to DLQ
026-08-01 20:42:32.459 - WARNING - Simulating Sensor Outage for AQI-S03: Injected Null AQI routed to DLQ
026-08-01 20:42:38.173 - WARNING - Simulating Sensor Outage for AQI-S01: Injected Null AQI routed to DLQ
026-08-01 20:42:41.182 - WARNING - Simulating Sensor Outage for AQI-S02: Injected Null AQI routed to DLQ
026-08-01 20:42:50.511 - WARNING - Simulating Sensor Outage for AQI-S01: Injected Null AQI routed to DLQ
026-08-01 20:42:56.527 - WARNING - Simulating Sensor Outage for AQI-S03: Injected Null AQI routed to DLQ
026-08-01 20:43:12.472 - WARNING - Simulating Sensor Outage for AQI-S02: Injected Null AQI routed to DLQ
026-08-01 20:43:26.008 - WARNING - Simulating Sensor Outage for AQI-S01: Injected Null AQI routed to DLQ
026-08-01 20:43:30.217 - WARNING - Simulating Sensor Outage for AQI-S03: Injected Null AQI routed to DLQ
026-08-01 20:43:33.823 - WARNING - Simulating Sensor Outage for AQI-S02: Injected Null AQI routed to DLQ
026-08-01 20:43:37.429 - WARNING - Simulating Sensor Outage for AQI-S03: Injected Null AQI routed to DLQ
026-08-01 20:43:45.748 - WARNING - Simulating Sensor Outage for AQI-S02: Injected Null AQI routed to DLQ
026-08-01 20:43:44.341 - WARNING - Simulating Sensor Outage for AQI-S02: Injected Null AQI routed to DLQ
026-08-01 20:43:47.647 - WARNING - Simulating Sensor Outage for AQI-S01: Injected Null AQI routed to DLQ
026-08-01 20:44:04.793 - WARNING - Simulating Sensor Outage for AQI-S01: Injected Null AQI routed to DLQ
026-08-01 20:44:06.298 - WARNING - Simulating Sensor Outage for AQI-S01: Injected Null AQI routed to DLQ
026-08-01 20:44:09.307 - WARNING - Simulating Sensor Outage for AQI-S03: Injected Null AQI routed to DLQ
026-08-01 20:44:18.628 - WARNING - Simulating Sensor Outage for AQI-S02: Injected Null AQI routed to DLQ
026-08-01 20:44:26.749 - WARNING - Simulating Sensor Outage for AQI-S01: Injected Null AQI routed to DLQ
026-08-01 20:44:35.476 - WARNING - Simulating Sensor Outage for AQI-S03: Injected Null AQI routed to DLQ
026-08-01 20:44:43.893 - WARNING - Simulating Sensor Outage for AQI-S03: Injected Null AQI routed to DLQ
026-08-01 20:44:44.795 - WARNING - Simulating Sensor Outage for AQI-S01: Injected Null AQI routed to DLQ
026-08-01 20:44:47.200 - WARNING - Simulating Sensor Outage for AQI-S03: Injected Null AQI routed to DLQ
026-08-01 20:44:47.802 - WARNING - Simulating Sensor Outage for AQI-S03: Injected Null AQI routed to DLQ
026-08-01 20:44:48.704 - WARNING - Simulating Sensor Outage for AQI-S02: Injected Null AQI routed to DLQ
026-08-01 20:44:50.809 - WARNING - Simulating Sensor Outage for AQI-S01: Injected Null AQI routed to DLQ
026-08-01 20:44:59.829 - WARNING - Simulating Sensor Outage for AQI-S01: Injected Null AQI routed to DLQ
026-08-01 20:45:10.052 - WARNING - Simulating Sensor Outage for AQI-S01: Injected Null AQI routed to DLQ
026-08-01 20:45:11.620 - WARNING - Simulating Sensor Outage for AQI-S02: Injected Null AQI routed to DLQ
026-08-01 20:45:18.835 - WARNING - Simulating Sensor Outage for AQI-S03: Injected Null AQI routed to DLQ
026-08-01 20:45:53.841 - WARNING - Simulating Sensor Outage for AQI-S01: Injected Null AQI routed to DLQ

```

